COLUMBIA UNIVERSITY
IN THE CITY OF NEW YORK

STAT 5703

Statistical Inference and Modeling for Data Science

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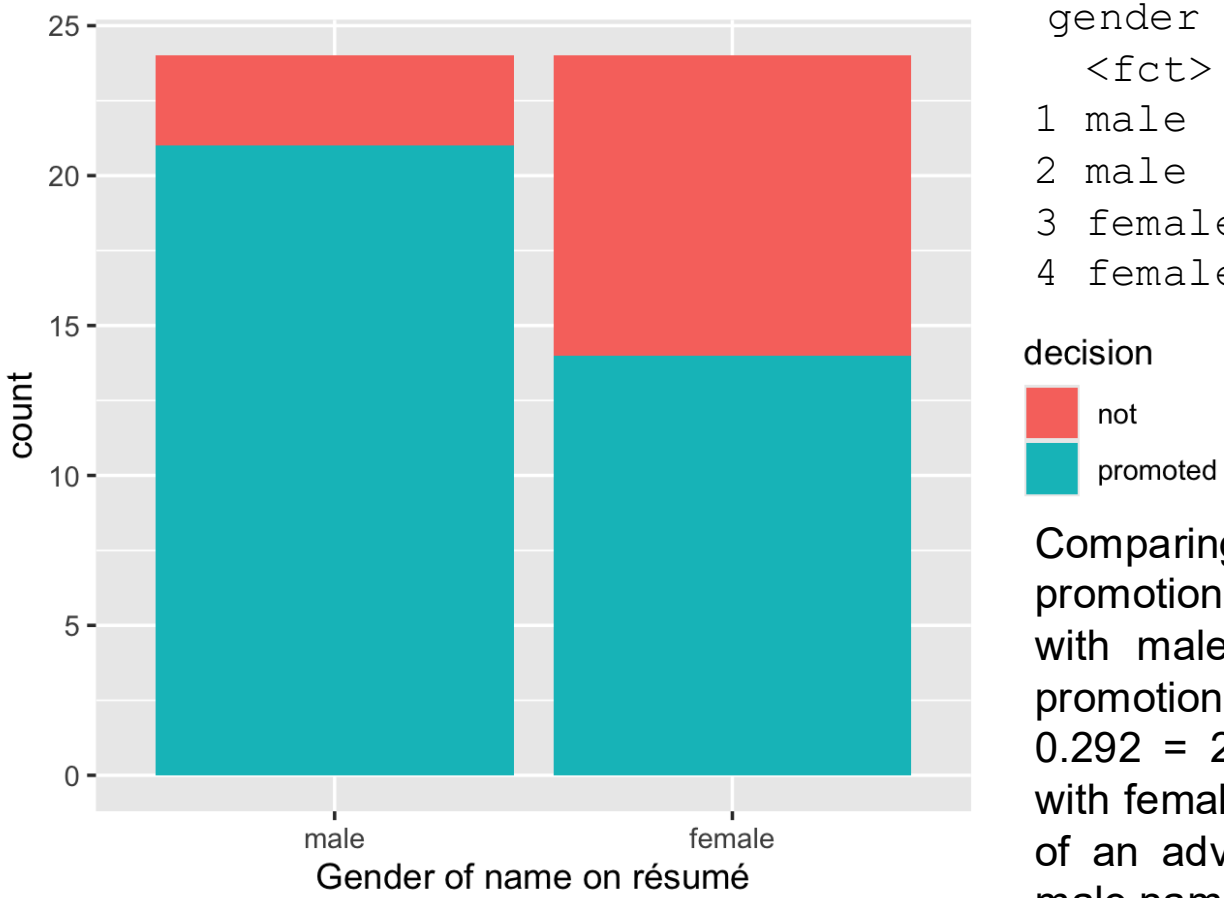
Hypothesis Tests

Now that we've studied confidence intervals and point estimators, let's study another commonly used method for statistical inference: *hypothesis testing*.

Motivating example: Does gender affect promotions at a bank?

- Say you are working at a bank in the 1970s and you are submitting your résumé to apply for a promotion. Will your gender affect your chances of getting promoted?
- To answer this question, we'll focus on data from a study published in the Journal of Applied Psychology in 1974.
- To begin the study, 48 bank supervisors were asked to assume the role of a hypothetical director of a bank with multiple branches. Every one of the bank supervisors was given a résumé and asked whether the candidate on the résumé was fit to be promoted to a new position in one of their branches.
- However, each of these 48 résumés were identical in all respects except one: the name of the applicant at the top of the résumé. Of the supervisors, 24 were randomly given résumés with stereotypically "male" names, while 24 of the supervisors were randomly given résumés with stereotypically "female" names.

Motivating example:
Does gender affect promotions at a bank?




```
 gender decision     n
   <fct>  <fct>    <int>
1 male   not          3
2 male   promoted    21
3 female not         10
4 female promoted    14
```

Comparing these two rates of promotion, it appears that résumés with male names were selected for promotion at a rate 0.875 - 0.583 = 0.292 = 29.2% higher than résumés with female names. This is suggestive of an advantage for résumés with a male name on it.

Stacked barplot relating gender to promotion decision.

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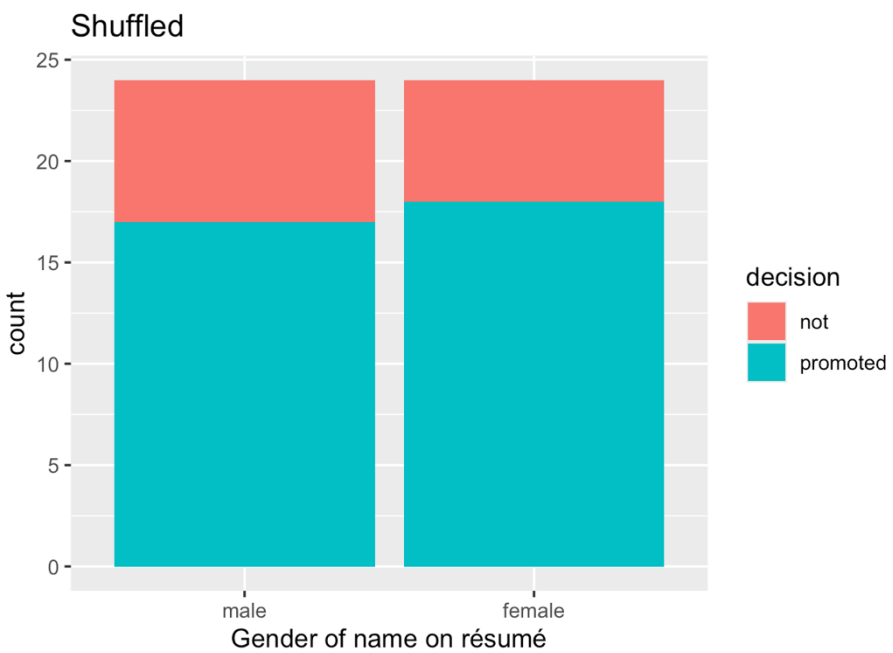
Q: Does this provide conclusive evidence that there is gender discrimination in promotions at banks?

A: One way to test this hypothesis is by reshuffling the sample.

```
  gender decision     n
    <fct>  <fct>    <int>
1 male   not           3
2 male   promoted     21
3 female not          10
4 female promoted     14
```

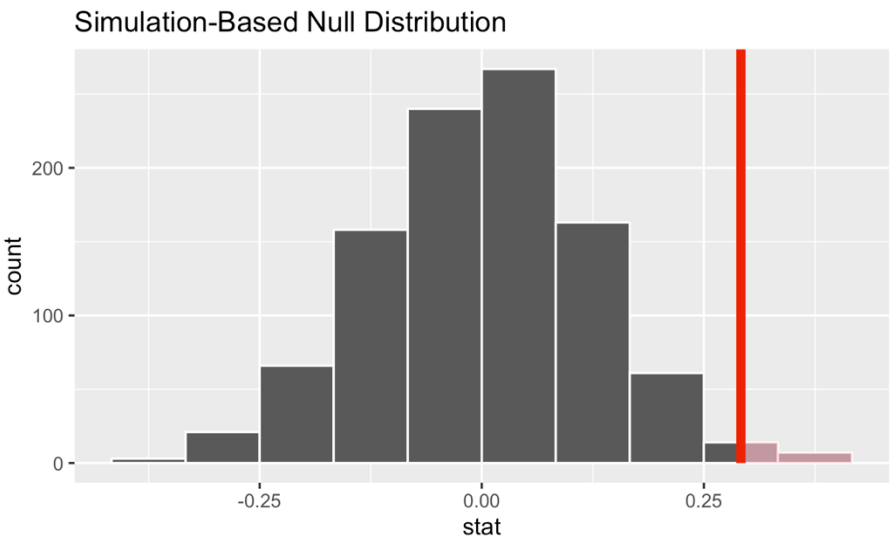
Motivating example: Does gender affect promotions at a bank?

- Try to imagine a hypothetical universe where no gender discrimination in promotions existed. In such a hypothetical universe, the gender of an applicant would have no bearing on their chances of promotion.
- Bringing things back to our promotions data frame, the gender variable would thus be an irrelevant label.
- If these gender labels were irrelevant, then we could randomly reassign them by “shuffling” them to no consequence!
- It appears the difference in “male names” versus “female names” promotion rates is now different.



Motivating example:
Does gender affect promotions at a bank?

- The final step is to measure how surprised we are by a promotion difference of 29.2% in a hypothesized universe of no gender discrimination. If the observed difference of 0.292 is highly unlikely, then we would be inclined to reject the validity of our hypothesized universe.
- This is known as a permutation test. You can use it to quantify the probability the original labeling was random or not by reshuffling many times.
- It is a *nonparametric* method. In this course we focus on *parametric* hypothesis testing.



Definition

- Hypothesis tests allow us to take a sample of data from a population and infer about the plausibility of competing hypotheses.
- A hypothesis is a claim about the value of the unknown population parameter.
- For example:

$$H_0: \mu \leq 0, H_a: \mu > 0$$

- There will always be two hypotheses; called *null* and *alternative*.
- Goal: Based on the sample, decide in favor of either H_0 or H_a

General Case

In general, we have:

$$H_0: \theta \in \Theta_0 \text{ vs. } H_a: \theta \in \Theta_1$$

where $\Theta_0 \cup \Theta_1$ is a partition of the parameter space Θ in the sense that

$$\Theta_0 \cup \Theta_1 = \Theta \ \& \ \Theta_0 \cap \Theta_1 = \emptyset$$

In the previous example, $H_0: \mu \leq 0, H_a: \mu > 0$

this means

$$\Theta = \mathbb{R}, \Theta_0 = (-\infty, 0], \Theta_1 = (0, \infty)$$

Decision

- Use a *test statistic* $T = T(X_1, \ldots, X_n) = T(\boldsymbol{X})$, where $\boldsymbol{X} = (X_1, \ldots, X_n)$
- Construct a *rejection region* (aka *critical region*) in the sample space, defined as:

$$R = \{(X_1, \ldots, X_n) : T(\boldsymbol{X}) > c\}$$

where c is called the *critical value*.

$$\boxed{\textbf{Rule}\text{: reject } H_0 \text{ if } \boldsymbol{X} \in R}$$

- Choose c such that the probability of rejecting a true null hypothesis does not exceed a small value, denoted α and called *significance level* of the test.

That is,

$$P(\boldsymbol{X} \in R | H_0) \leq \alpha$$

Power Function

More generally, define the *power function*

$$\boxed{\pi(\theta) = P(\boldsymbol{X} \in R|\theta)}$$

which is a function that gives, for any θ, the probability of rejecting the null hypothesis when the true parameter is equal to θ.

Properties (see Example 1 for a graph):

- When $\theta \in \Theta_0$, then

$$\pi(\theta) = \text{Type I error} \leq \alpha = \max_{\theta \in \Theta_0} \pi(\theta)$$

- When $\theta \in \Theta_1$, Type II error = $\beta(\theta) = 1 - \pi(\theta)$

Example 1:

Let $X_1, \ldots, X_n \sim N(\mu, \sigma)$ where σ is known.

Test: $H_0: \mu \leq 0,\ H_a: \mu > 0$

Use the following test statistic:

$$T(X_1, \ldots, X_n) = \overline{X}$$

Then the rejection region is:

$$R = \left\{(X_1, \ldots, X_n): \overline{X} > c\right\}$$

Find:

a) the formula for the power function $\pi(\mu)$

b) the value of c such that $\alpha = 0.05$

Example 1 (Solution):

$$\pi(\mu) = P(\overline{x} > c|\mu) = P\left(\frac{\overline{x} - \mu}{\frac{\sigma}{\sqrt{n}}} > \frac{c - \mu}{\frac{\sigma}{\sqrt{n}}}|\mu\right)$$

$$= P\left(Z > \frac{\sqrt{n}(c - \mu)}{\sigma}\right) = 1 - \Phi\left(\frac{\sqrt{n}(c - \mu)}{\sigma}\right)$$

Where Φ is the notation of the cdf of a standard normal $N(0, 1)$ random variable.

Solution

Power function: $\pi(\mu) = 1 - \Phi\left(\frac{\sqrt{n}(c-\mu)}{\sigma}\right)$

We want c such that $\max_{\mu \in H_0} \pi(\mu) = 0.05$

or $1 - \Phi\left(\frac{\sqrt{n}(c-\mu)}{\sigma}\right) \leq 0.05$ when $\mu \leq 0$

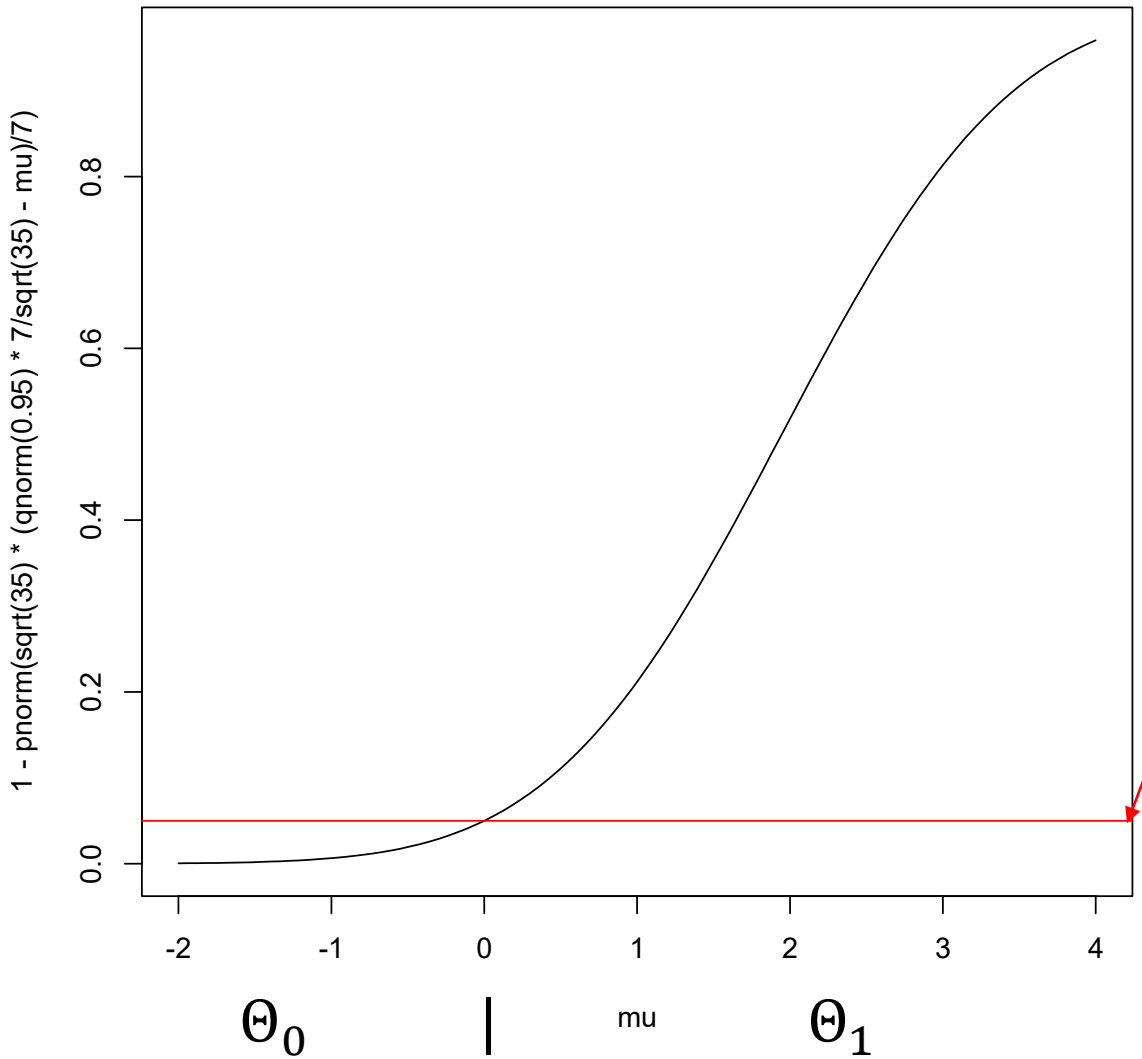
$1 - \Phi\left(\frac{\sqrt{n}(c-\mu)}{\sigma}\right)$ is increasing in μ (derivative is >0)

$\Rightarrow$ max is at $\mu = 0$

$\Rightarrow 1 - \Phi\left(\frac{\sqrt{n}(c-0)}{\sigma}\right) = 0.05 \Leftrightarrow \Phi\left(\frac{\sqrt{n}(c-0)}{\sigma}\right) = 0.95$

$\Rightarrow \frac{\sqrt{n}\,c}{\sigma} = 1.645 \Rightarrow \boxed{c = 1.645\frac{\sigma}{\sqrt{n}}}$

Example 1: Power Curve



$n = 35$

$\sigma = 7$

$\alpha = 0.05$

Example 1: Solution

To test $H_0: \mu \le 0$ vs. $H_a: \mu > 0$

Use Rule to reject H_0 when

$$\bar{X} > 1.645 \frac{\sigma}{\sqrt{n}} \quad \text{when } \alpha = 0.05$$

$$\Leftrightarrow \quad \frac{\bar{X}}{\sigma/\sqrt{n}} > 1.645$$

$\longrightarrow$ z-test

$$\Leftrightarrow \quad \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} > z_{crit}, \quad \alpha = 0.05$$

Exercise 1

Let $X_1, X_2, \ldots, X_n$ be a random sample from the population random variable $X \sim \mathrm{N}(\mu, \sigma)$.

Test:

$$\mathrm{H}_0: \mu \geq 75 \text{ vs. } \mathrm{H}_a: \mu < 75$$

using the rejection region

$$R = \{\bar{X} < 70.8\}$$

with $n = 25$, $\sigma = 9$

a) Find α

b) Find $\beta(72)$

c) Draw the power curve (using software)

Power curve

- β = P(Type II error) is the conditional probability that we fail to reject H_0 when H_a is true;
- 1 – β = power of the test is the conditional probability that we reject H_0 when H_a is true;
- That is, the graph of 1 – β vs. the parameter values is called a *power curve*.
- For *z*-test $\beta(\mu)$ can be calculated by observing

$$\frac{\bar{x}-\mu_0}{\frac{\sigma}{\sqrt{n}}} \sim N\left(\frac{\mu-\mu_0}{\frac{\sigma}{\sqrt{n}}}, 1\right)$$

Example 2

Suppose we have *one* observation:

$$X \sim \text{Binomial}(20, p)$$

Test: H_0: $p = 0.25$ vs. H_a: $p > 0.25$

Using the rejection region

$$R = \{X \geq 8\}$$

a) Find α and $\beta(0.3)$

b) Repeat a) for the rejection region

$$R = \{X \geq 9\}$$

Example 2: Solution

Power function: $\pi(p) = P(R|p) = P(X \geq 8 \mid p)$

$X \sim \mathrm{Bin}(20, p) \Rightarrow \pi(p) = p(X = 8) + p(X = 9) + \cdots + \mathrm{P}(X = 20)$

$$= \binom{20}{8} p^8 (1-p)^{12} + \ldots + \binom{20}{20} p^{20} (1-p)^0$$

$$= \sum_{x=8}^{20} \binom{20}{x} p^x (1-p)^{20-x}$$

$$\alpha = \max_{p \in H_0} \pi(p) = \pi(0.25) = \sum_{x=8}^{20} \binom{20}{x} 0.25^x (0.75)^{20-x}$$

$$= 0.1019 = 1 - \mathrm{pbinom}(7, 20, .25)$$

Example 2: Solution

To find the Type II error β we need p from H_a. That is, $p = 0.3$

$$\pi(0.3) = \sum_{x=8}^{20} \binom{20}{x} (0.3)^x (0.7)^{20-x} = 0.2277$$

$\Rightarrow$ Type II error $= 1 - \pi(0.3) = 0.7723 = \beta(0.3)$

$\beta(0.5) = 0.1316$

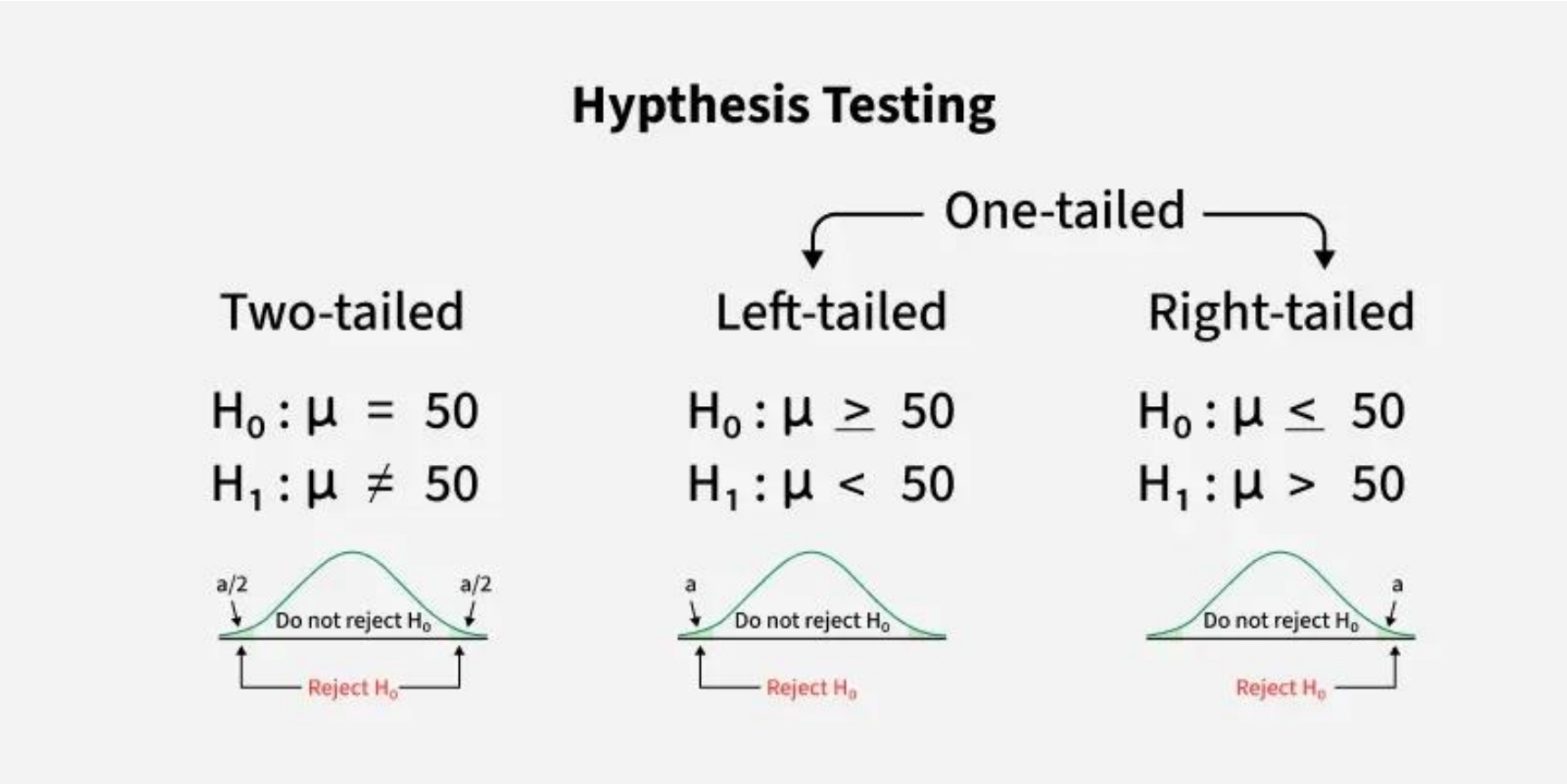
$R = \{x \geq 9\} \Rightarrow \pi(p) = \sum_{x=9}^{20} \binom{20}{x} p^x (1-p)^{20-x}$

$\Rightarrow \alpha = \pi(0.25) = 0.0409$

$\beta(0.3) = 0.8867$, $p(0.5) = 0.2517$

Tests about the population mean μ

There are many versions of theses tests, but they are all based on a similar principle:



Tests about the population mean μ

<u>Case 1</u>: Sample size is large ($n \geq 30$)

Test statistic:

$$Z = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$$

When n is large $Z \xrightarrow{d} N(0, 1)$

	Left-tail	**Two-tail**	**Right-tail**
H_0	$\mu \geq \mu_0$	$\mu = \mu_0$	$\mu \leq \mu_0$
H_a	$\mu < \mu_0$	$\mu \neq \mu_0$	$\mu > \mu_0$
RR	$Z < -z_\alpha$	$\lvert Z \rvert > z_{\alpha/2}$	$Z > z_\alpha$

Tests about the population mean μ

Case 2: Normal population and σ is *known*

(this is the case we started with)

Test statistic:

$$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$$

Then $Z \sim N(0, 1)$

	Left-tail	**Two-tail**	**Right-tail**
H_0	$\mu \geq \mu_0$	$\mu = \mu_0$	$\mu \leq \mu_0$
H_a	$\mu < \mu_0$	$\mu \neq \mu_0$	$\mu > \mu_0$
RR	$Z < -z_\alpha$	$\lvert Z \rvert > z_{\alpha/2}$	$Z > z_\alpha$

Tests about the population mean μ

Case 3: Normal population and σ is *unknown,* aka small sample *t*-test.

$$T = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

$$T \sim t_{n\text{-}1}$$

Assumption: $X_1, X_2, \ldots, X_n \sim \mathrm{N}(\mu, \sigma^2)$

	Left-tail	**Two-tail**	**Right-tail**
H_0	$\mu \geq \mu_0$	$\mu = \mu_0$	$\mu \leq \mu_0$
H_a	$\mu < \mu_0$	$\mu \neq \mu_0$	$\mu > \mu_0$
RR	$T < -t_\alpha$	$\lvert T \rvert > t_{\alpha/2}$	$T > t_\alpha$

Example 3

Test the hypotheses:

$$H_0: \mu \leq 15 \text{ vs. } H_a: \mu > 15$$

Given data summary:

$\bar{X} = 17.986, s = 5.937, n = 70, \alpha = 0.05$

a) Calculate the test statistic.

b) Draw a conclusion.

Example 3:

Test the hypotheses:

H_0: $\mu \leq 15$ vs. H_a: $\mu > 15$

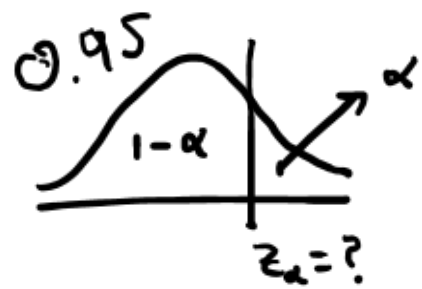
Given

$\bar{X} = 17.986, s = 5.937, n = 70, \alpha = 0.05$

T.S. $z = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}} = \dfrac{17.986 - 15}{\dfrac{5.937}{\sqrt{70}}} = 4.21$

Critical $z_\alpha = z_{0.05}$

$= 1.645$



Conclusion:

Since

$4.21 > 1.645$

reject H_0

Exercise 2:

Consider testing the hypotheses

$$H_0: \mu = \mu_0 \text{ vs. } H_a: \mu \neq \mu_0$$

Given a random sample $X_1, \ldots, X_n \sim N(\mu, \sigma)$ where σ is known.

Derive the formula for the power of the rejection rule:

$$\sqrt{n}\left|\frac{\bar{x} - \mu_0}{\sigma}\right| > Z_{\frac{\alpha}{2}}$$

Sketch the graph when $\alpha = 0.05$, $n = 58$, $\sigma = 28$, $\bar{x} = 91, \mu_0 = 90$

Exercise

Test the hypotheses:

H_0: $\mu \leq 25$ vs. H_a: $\mu > 25$

Given

$$\bar{X} = 27.54, s = 5.47, n = 5, \alpha = 0.05$$

Calculate the test statistic and draw a conclusion.

Example 5:

Test the hypotheses:

H_0: $\mu \leq 25$ vs. H_a: $\mu > 25$

Given

$\bar{X} = 27.54, s = 5.47, n = 5, \alpha = 0.05$

$$t = \frac{27.54 - 25}{\frac{5.47}{\sqrt{5}}} = 1.04, \quad t_{0.05} = 2.132$$

Conclusion: Since $1.04 \not> 2.132$, do not reject H_0

Power of the Test

Note: Calculating $\beta(\mu)$ for *t*-test can be done only numerically!

In R: **power.t.test**

For large sample *z*-test:

$\beta(\mu)$ can be calculated from observing

$$\frac{\overline{x}-\mu_0}{\frac{s}{\sqrt{n}}} \approx N\left(\frac{\mu-\mu_0}{\frac{s}{\sqrt{n}}}, 1\right)$$

Sample Variance and the Chi-squared Distribution

Definition:

If $Z_1, Z_2, \ldots, Z_n$ are *independent* random variables each having a *standard normal* distribution then

$$U = Z_1^2 + Z_2^2 + \cdots + Z_n^2$$

has a chi-squared distribution with *n degrees of freedom*, written as $U \sim \chi_n^2$

Statistics that have the Chi-squared distribution:

1. The statistic used to detect independence between two categorical variables

$df = (r - 1)(c - 1)$

2. Likelihood ratio tests

We will go over this topic next week.

3. The sample variance s^2

Theorem: Let $X_1, X_2, \ldots, X_n$ denote a random sample from a normal distribution with mean μ and standard deviation σ. Recall sample variance

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

and define

$$U = \sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{\sigma}\right)^2 = \frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1}$$

Then $U \sim \chi^2_{n-1}$

That is, U has a chi-square distribution with $df = n - 1$

Proof:

Start by assuming μ is *known*.

Then, since each $X_i \sim \mathrm{N}(\mu, \sigma)$,

$$Z_i = \frac{X_i - \mu}{\sigma} \sim N(0,1), \qquad i = 1, \ldots, n$$

$$\Rightarrow \left(\frac{X_1 - \mu}{\sigma}\right)^2 + \ldots + \left(\frac{X_n - \mu}{\sigma}\right)^2 \sim \chi_n^2$$

$$\Rightarrow \quad \frac{\sum (x_i - \mu)^2}{\sigma^2} \sim \chi_n^2$$

We want to show $\frac{\sum (x_i - \bar{x})^2}{\sigma^2} \sim \chi_{n-1}^2$

Proof continued: μ is unknown

$$\sum_{i=1}^{n}(x_i-\mu)^2=\sum_{i=1}^{n}(x_i-\bar{x}+\bar{x}-\mu)^2=$$

$$=\sum_{i=1}^{n}(x_i-\bar{x})^2+2\sum_{i=1}^{n}(x_i-\bar{x})(\bar{x}-\mu)+\sum_{i=1}^{n}(\bar{x}-\mu)^2$$

But

$$\sum_{i=1}^{n}(x_i-\bar{x})(\bar{x}-\mu)=(\bar{x}-\mu)\sum_{i=1}^{n}(x_i-\bar{x})$$

$$=(\bar{x}-\mu)\left(\sum_{i=1}^{n}x_i-\sum_{i=1}^{n}\bar{x}\right)=(\bar{x}-\mu)\left(\sum x_i-n\bar{x}\right)$$

$$=(\bar{x}-\mu)\left(\sum x_i-\sum x_i\right)=(\bar{x}-\mu)\times 0=0$$

Proof:

From previous slide we got the algebraic identity:

$$\sum_{i=1}^{n}(x_i-\mu)^2=\sum_{i=1}^{n}(x_i-\bar{x})^2+\sum_{i=1}^{n}(\bar{x}-\mu)^2$$

Therefore,

$$\underbrace{\sum_{i=1}^{n}(X_i-\mu)^2/\sigma^2}_{\substack{\sim\chi_n^2\\ \text{(Why?)}}}=\underbrace{\sum_{i=1}^{n}(X_i-\bar{X})^2/\sigma^2}_{=A\sim ?}+\underbrace{n(\bar{X}-\mu)^2/\sigma^2}_{\substack{=B\sim\chi_1^2\\ \text{(Why?)}}}$$

Proof (continued):

$$\underbrace{\sum_{i=1}^{n}(X_i-\mu)^2/\sigma^2}_{\sim\chi_n^2}=\underbrace{\sum_{i=1}^{n}(X_i-\bar{X})^2/\sigma^2}_{=A\sim ?}+\underbrace{n(\bar{X}-\mu)^2/\sigma^2}_{=B\sim\chi_1^2}$$

$$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}\sim N(0,1)\Rightarrow B=\left(\frac{\bar{X}-\mu}{\frac{\sigma}{\sqrt{n}}}\right)^2\sim\chi_1^2$$

$$\frac{X_i-\mu}{\sigma}\sim N(0,1)\Rightarrow\left(\frac{X_i-\mu}{\sigma}\right)^2\sim Z^2=\chi_1^2,\forall i=1,\dots,n$$

<u>Aside:</u> Cochran's Theorem (loosely presented)

If A and B are independent and $A+B\sim\chi_{n_1+n_2}^2$, and $B\sim\chi_{n_2}^2$, then $A\sim\chi_{n_1}^2$

$$\Rightarrow A\sim X_{n-1}^2$$

3. Sample variance

Main result: If the sample is normal, then:

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$$

Corollary: Recall that $E(\chi^2_\nu) = \nu$

$$\Rightarrow \quad E\left[\frac{(n-1)S^2}{\sigma^2}\right] = E(\chi^2_{n-1}) = n-1$$

$$\Leftrightarrow \quad \frac{\cancel{(n-1)}\,E(S^2)}{\sigma^2} = \cancel{n-1} \Leftrightarrow \boxed{E(S^2) = \sigma^2}$$

$\Rightarrow$ S^2 is unbiased for σ^2

Example 6

Suppose that $X_1, X_2, \ldots, X_{10}$ is a sample of size n = 10 from a normal distribution with mean μ =100 and standard deviation σ = 15.

Recall that

$$S = \sqrt{\frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}}$$

is the sample standard deviation.

Find P(10 < S < 20).

Note in this case the chi-squared statistic is equal to: $= \dfrac{(n-1)s^2}{\sigma^2} = \dfrac{(9)s^2}{(15)^2}$

And has a chi-square distribution with

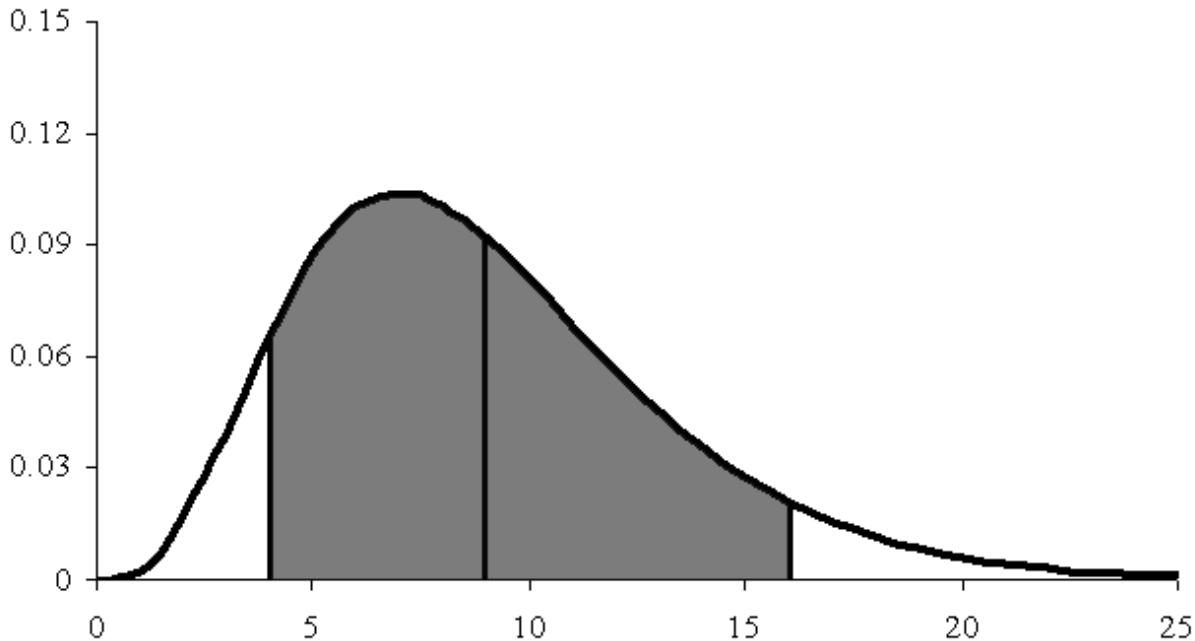
$df = n - 1 = 9$

$$P\left[10 < s < 20\right] = P\left[100 < s^2 < 400\right]$$

$$= P\left[\frac{9(100)}{15^2} < \frac{9s^2}{15^2} < \frac{9(400)}{15^2}\right]$$

$$= P\left[4 < U < 16\right]$$

The `R` function `pchisq(x,df)` computes the *left* areas $P\left(\chi^2_{df} < x\right)$



$P\,(4 < U < 16) =$ `pchisq(16, 9) - pchisq(4, 9)`

$= \quad 0.9331184 - \ 0.08858747 = 0.8445309$

Summary

Let $X_1, X_2, \ldots, X_n$ be a random sample from the population $N(\mu, \sigma)$

Then:

$$\boxed{\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}}$$

Example 6

Let $X_1, X_2, \ldots, X_n$ be a random sample from the population random variable $X \sim \mathrm{N}(\mu, \sigma)$

Test: H_0: $\sigma \leq 5$ vs. H_a: $\sigma > 5$

Using the rejection region

$$R = \left\{ \sum_{i=1}^{n} (X_i - \bar{X})^2 > c \right\}$$

With $n = 45$, $s = 6.42$ and $\alpha = 0.05$

a) Find c;

b) State the conclusion.

Example 6 Solution

We have that:

$$\frac{(n-1)s^2}{\sigma^2} \sim X_{n-1}^2, \text{where } s^2 = \sum \frac{(x_i - \bar{x})^2}{n-1}$$

$$\Rightarrow \pi(\sigma) = P(R|\sigma) = \mathrm{P}(\textstyle\sum_{i=1}^{n}(X_i - \bar{X})^2 > c|\sigma)$$

$$= P\left(\frac{(n-1)S^2}{\sigma^2} > \frac{c}{\sigma^2}\right) = P\left(X_{n-1}^2 > \frac{c}{\sigma^2}\right)$$

$$= 1 - F_{\chi_{n-1}^2}\left(\frac{c}{\sigma^2}\right)$$

where $F_{\chi_{n-1}^2}$ is the cdf of the χ_{n-1}^2 distribution.

Note that $\pi'(\sigma) = -f_{\chi_{n-1}^2}\left(\frac{c}{\sigma^2}\right)(-2c)\sigma^{-3} = \frac{2c}{\sigma} f_{\chi_{n-1}^2}\left(\frac{c}{\sigma^2}\right) > 0$

Therefore, $\pi(\sigma)$ is an increasing function.

Example 6 Solution

$$\Rightarrow \max_{H_0} \pi(\sigma) = \max_{\sigma \le 5} \pi(\sigma) = \pi(5)$$

$$\Rightarrow \alpha = \pi(5) \Leftrightarrow 0.05 = 1 - F_{\chi^2_{44}}\left(\frac{c}{25}\right)$$

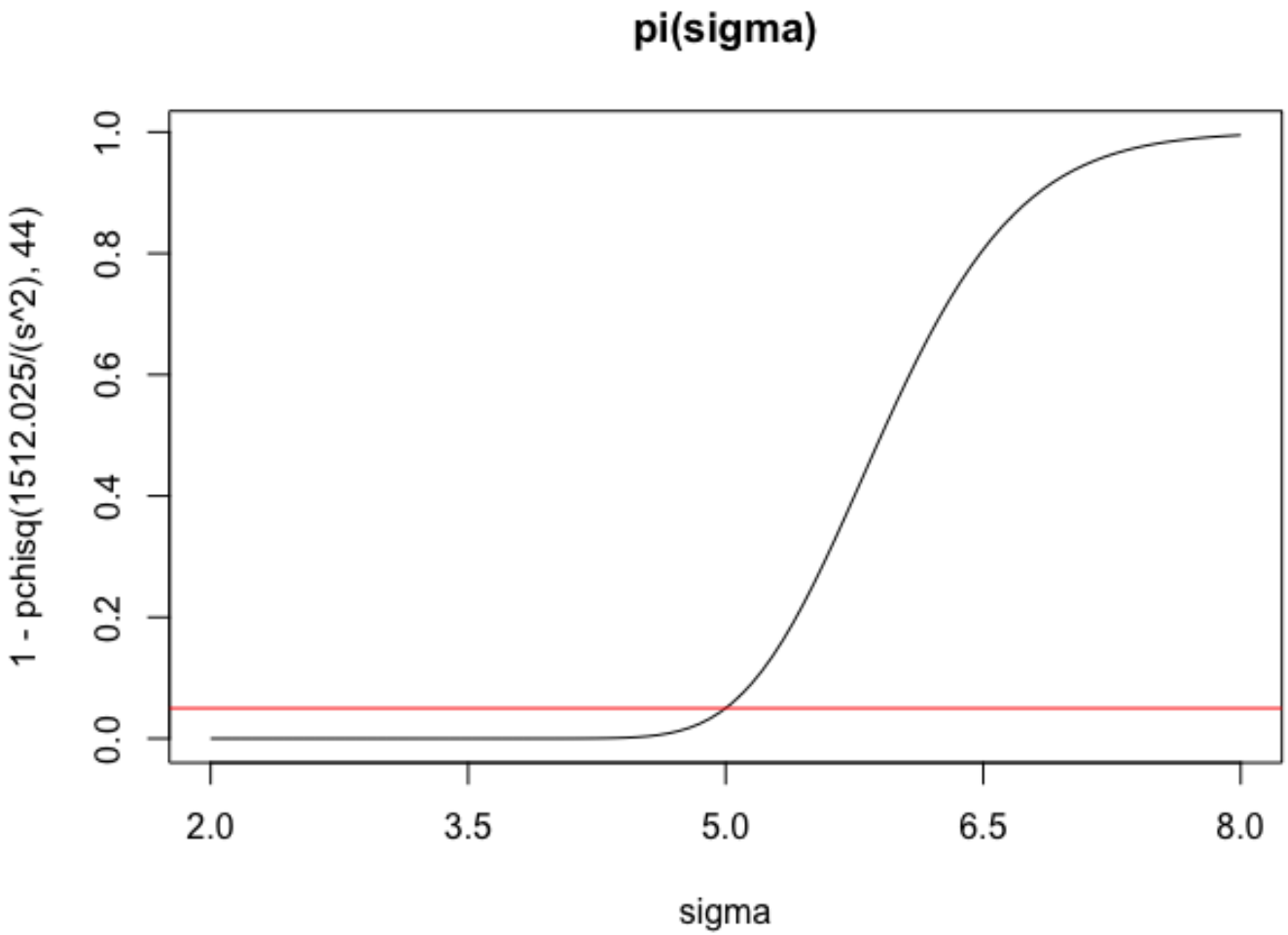
$$\Leftrightarrow F_{\chi^2_{44}}\left(\frac{c}{25}\right) = 0.95$$

$$\Leftrightarrow \frac{c}{25} = 95^{th} \text{ quatile of } \chi^2_{44} = 60.481$$

(from table)

$$\Rightarrow c = 25(60.481) = \boxed{1512.025}$$

b) $\sum (x_i - \bar{x})^2 = (n-1)s^2 = 44(6.42)^2 = 1813.52$

Since $1813.52 > 1512.025$ we reject H_0

Example 1: Graph of $\pi(\sigma)$